Module 6 — HIJING Standalone

Supplementary to the sPHENIX Standalone Course

1. HIJING in one paragraph

HIJING (Heavy-Ion Jet INteraction Generator) combines PYTHIA-like hard scattering with a Glauber model of soft nucleon-nucleon collisions. Nuclei are treated as collections of nucleons with Woods-Saxon density. Hard processes use lowest-order pQCD with a minimum p_T cutoff. Nuclear effects include EKS98-like shadowing and (optionally) jet quenching (dE/dx ≈ 1 GeV/fm).

2. Build recipe

```
cd $HOME/src
# obtain hijing-1.383 tarball (RHIC MC repository or mirror)
tar xf hijing-1.383.tgz
cd hijing-1.383
gfortran -c -O2 -fno-automatic -std=legacy *.f
ar rcs libhijing.a *.o
```

Note: The -fno-automatic flag is essential — HIJING's COMMON-block design assumes static storage. Without it, some arrays end up on the stack and get corrupted between calls.

3. Minimal driver (hmain.f)

```
      PROGRAM HMAIN
      IMPLICIT NONE
      REAL*4 EFRM, BMIN, BMAX
      INTEGER IAP, IZP, IAT, IZT, N, J
      CHARACTER*4 FRAME, PROJ, TARG
      COMMON/HIMAIN1/ NATT
      INTEGER NATT
      EFRM = 200.0
      FRAME = 'CMS'
      PROJ  = 'A'
```

```
TARG  = 'A'
IAP = 197; IZP = 79    ! Au
IAT = 197; IZT = 79
BMIN = 0.0; BMAX = 15.0
CALL HIJSET(EFRM, FRAME, PROJ, TARG, IAP, IZP, IAT, IZT)
DO N=1, 10
   CALL HIJING(FRAME, BMIN, BMAX)
   WRITE(*,*) 'Event', N, ' multiplicity =', NATT
ENDDO
END
```

4. Key COMMON blocks

	Holds	What you care about
	NATT	Total multiplicity of this event
	KATT(4,N), PATT(4,N)	Particle list (PDG ID, status, parents; momenta)
	HINT1(100), IHPR2(50)	Knobs: $\sqrt{s}$, Nbin, jet quenching toggle, etc.
	YP(3,300), YT(3,300)	Nucleon coordinates for projectile/target
	NPJ, KPJT, PJPX,Y,Z,E	Projectile jets
	NTJ, KFTJ, PJTX,Y,Z,E	Target jets

5. Useful IHPR2 toggles

	Meaning	Default
	Jet quenching on/off	1 (on)
	Nuclear shadowing on/off	1 (on)
	Maximum number of MPI per NN	10
	QCD branching off/on	1 (on)
	Warn on problem events	1
	Soft interaction on/off	1
	Decay K0s and Λ	1 (decay)
	Gluon vs quark shadowing flag	1

6. Impact parameter $\rightarrow$ centrality

HIJING samples b uniformly in the interval [BMIN, BMAX]. For centrality classes, run the generator with wide b range and later cut on Σ E_T (or N_part) to select a centrality window.

Centrality	Approx b / R (Au+Au @ 200)
0–10%	b < 4.7 fm
10–20%	4.7 – 6.6 fm
20–30%	6.6 – 8.1 fm
30–40%	8.1 – 9.4 fm
50–60%	10.5 – 11.6 fm
70–80%	12.4 – 13.3 fm

7. ASCII output pattern

The simplest way to use HIJING output is a human-readable ASCII dump in your driver, then a Python / ROOT parser. Dump loop:

```
      DO J=1, NATT
         WRITE(10,'(I5,I12,4F12.5)') J, KATT(J,1),
     &        PATT(J,1),PATT(J,2),PATT(J,3),PATT(J,4)
      ENDDO
```

Where KATT(J,1) is the PDG ID (or HIJING's internal code, depending on version) and PATT(J,1..4) are px, py, pz, E.

8. Linking your driver

```
gfortran hmain.f -L. -lhijing -o hmain
./hmain    # writes event summaries to stdout
```

9. Troubleshooting

Symptom	Cause / fix
NATT = 0 for every event	PROJ/TARG mismatch, or check LUN for events
Stack overflow on large A*A	Rebuild with <code>-fno-automatic</code>
Different NATT each run for same seed	HIJING seed set through <code>HIRND</code>; set it explicitly
Crash in <code>HIJFRG</code>	Fragmentation failure, usually harmless — skip event

10. A sanity plot

If you dump Au+Au @ 200 GeV, minimum bias, you should see charged $dN/d\eta|_{\eta=0}$ averaging around 175 for the top 5% centrality class — consistent with PHENIX measurements. If you see 30 or 700, your centrality definition or acceptance is off.